

Ali Alemami

Systems & Backend Developer

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PROFILE

Systems and backend developer with hands-on experience building Unix shells, HTTP servers, and concurrent simulations in C/C++ at 42 Amman, alongside enterprise desktop applications in C#/.NET. CS student at Hashemite University, 22/24 projects into the 42 Common Core.

EDUCATION

42 Amman 2025 - Present

Common Core - 22/24 Projects Complete

- Conducted 50+ peer code reviews and defense audits, evaluating memory leaks (Valgrind), race conditions, and architecture.

The Hashemite University 2022 - 2026 (Expected)

Bachelor of Science in Computer Science

- Peer

- Cour

TECHNICAL SKILLS

Languages: C (POSIX/C99), C++ (98/11/17), C# (.NET), SQL (T-SQL), Bash

Systems:

FEATURED PROJECTS

Philosophers (github.com/ali-alemami/philos)

C / Concurrency

Race-free multithreaded simulation of Dijkstra's Dining Philosophers using POSIX threads, mutex locks, asynchronous monitors, and microsecond delta timers.

Webserv (In Progress) (github.com/ali-alemami/webserv)

C++98 / Networking

High-performance HTTP/1.1 server using non-blocking epoll I/O multiplexing, stateful request parsing (GET/POST/DELETE), NGINX-style configs, and CGI execution.

Inception (github.com/ali-alemami/inception)

DevOps / Docker

Multi-service infrastructure: NGINX (TLS v1.2/v1.3), WordPress (PHP-FPM), and MariaDB microservices via Docker Compose on Alpine Linux.

Minishell (github.com/ali-alemami/minishell)

C / Unix Systems

POSIX-compliant Unix shell with AST parser, subshell execution, pipelines, heredocs, variable expansions, and signal handling.

mini_rt (github.com/ali-alemami/mini_rt)

C / Graphics

3D raytracing engine from scratch: vector math, Phong illumination (diffuse, ambient, specular, shadows), and geometric intersections.

Bank & ATM System (github.com/ali-alemami/Bank-and-ATM-System-CPP)

C++ / OOP

Modular dual-engine banking application with bitmask-based access controls, custom file serialization (###), transaction audit logs, and file-based persistence.

Contacts Management App (github.com/ali-alemami/Contacts-App-3Tier)

C# / .NET / SQL

3-tier desktop application with strict layer separation (Presentation, Business, Data Access), parameterized T-SQL queries via ADO.NET.

Push Swap (github.com/ali-alemami/push_swap)

C / Algorithms

Sorting algorithm using two stacks and bitwise radix operations to sort arbitrary integers in minimal instruction counts.

MASTER REPOSITORIES

42 Core Archive (github.com/ali-alemami/42-core)

All 42 common core projects: Unix shells, 3D raytracers, C++ modules 00-09. 22/24 projects complete.

Programming Advices Core (github.com/ali-alemami/Programming-Advices-Core)

225+ algorithmic solutions across 5 levels, 5 relational database schemas, and 3-tier desktop architecture in C#/NET.